

Direct variation



- 1 If $(4, 512)$ is on the curve, $P = kQ^4$, solve for k .
- 2 A variable M is directly proportional to the square of N .
 - a Using k as the constant of proportionality, form an equation for M in terms of N .
 - b Given that $(4, 64)$ satisfies the equation, solve for k .
- 3 Consider the relationship where y is directly proportional to the cube of x .
 - a Using k as a constant of proportionality, state the equation relating x and y .
 - b The following table of values shows the relationship between x and y :

x	1	2	3	4	5
y	3	24	81	192	375

Solve for k , the constant of proportionality.

- 4 y is a power function which is directly proportional to x^n .
 - a Using k as the proportionality constant, form an equation for y in terms of x .
 - b The table below shows values that satisfy the relationship $y = kx^n$, for the case when $n = 2$.

x	2	4	6	8	10
y	32	128	288	512	800

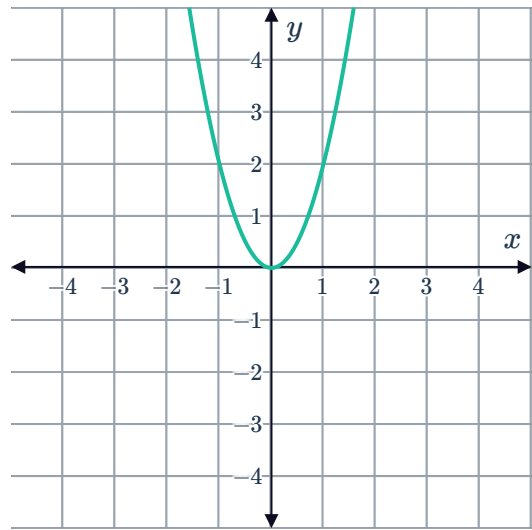
Solve for the constant of proportionality k . Round your answer to the nearest integer.

- 5 y is a power function which is directly proportional to x^n .
 - a Using k as the proportionality constant, form an equation for y in terms of x .
 - b The table below shows values that satisfy the relationship $y = kx^n$, for the case when $n = 3$.

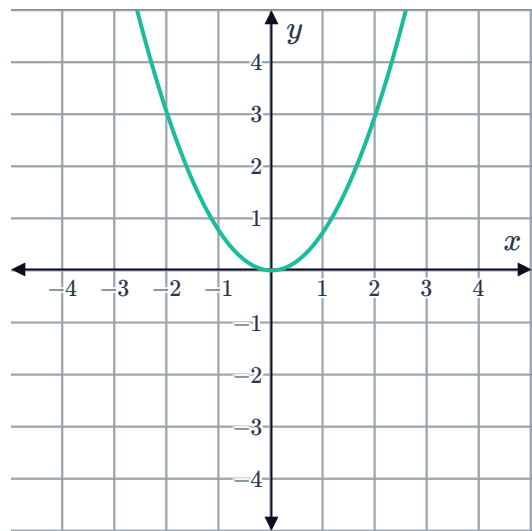
x	2	4	6	8	10
y	-48	-384	-1296	-3072	-6000

Solve for the constant of proportionality k . Round your answer to the nearest integer.

- 6 For the quadratic function pictured here, determine the constant of proportionality k :

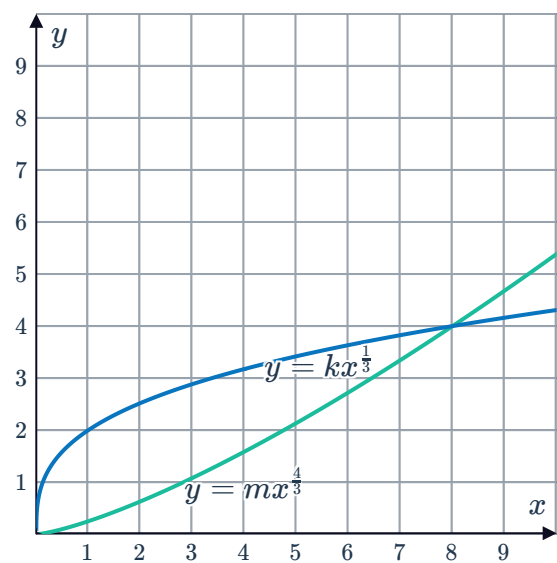


- 7 For the quadratic shown, determine the constant of proportionality k :



- 8 The following graph shows the intersection of the two functions $y = mx^{\frac{4}{3}}$ and $y = kx^{\frac{1}{3}}$:

- Find the value of m .
- Find the value of k .





Inverse variation

- 9 If y varies inversely with x , write an equation that uses k as the constant of variation.
- 10 If $(10, 15)$ is on the curve $P = \frac{k}{Q}$, solve for k .
- 11 Determine whether each of the following is an example of a direct variation or an inverse variation:
- a The variation relating the distance between two locations on a map and the actual distance between the two locations.
 - b The variation relating the number of workers hired to build a house and the time required to build the house.
- 12 If r varies inversely with a , write an equation that uses k as the constant of variation.
- 13 Determine whether the following equations represent an inverse relationship between x and y :
- a $x = 1 + y^3$
 - b $x = \frac{8}{y^2}$
 - c $y = 6x + 8$
 - d $xy = -7$
 - e $x = \frac{2}{y}$
 - f $xy = 5x$
- 14 Consider the equation $s = \frac{375}{t}$.
- a State the constant of proportionality.
 - b Find the exact value of s when $t = 6$.
 - c Find the exact value of s when $t = 12$.
- 15 State whether the following tables represent an inversely proportional relationship between x and y .

a

x	1	2	3	4
y	3	1.5	1	0.75

b

x	1	2	3	4
y	36	18	12	9

c

x	1	5	6	10
y	3	75	108	300

d

x	1	2	3	4
y	4	5	6	7



- 16 m is proportional to $\frac{1}{p}$. Consider the values in the table which represents the relationship.

p	4	6	7	x
m	63	y	36	28

- a Determine the constant of proportionality, k .
b Find the values of x and y .

- 17 Find the equation relating t and s for each of the following tables of values:

a

s	1	2	3	4
t	48	24	16	12

b

s	3	6	9	12
t	$\frac{2}{9}$	$\frac{1}{18}$	$\frac{2}{81}$	$\frac{1}{72}$

- 18 Use technology to graph the inverse relationship $y = \frac{6}{x}$.

- a How many x -intercepts does the graph have?
b How many y -intercepts does the graph have?

Applications

- 19 In exchange rate calculations the constant of proportionality is the exchange rate itself. Determine the exchange rate, k , if someone can purchase:

- a 889.00 AUD with 700.00 USD. b 553.00 USD with 700.00 AUD.

- 20 The heart mass H of a mammal is proportional to its body mass M .

- a Using k as the constant of proportionality, form an equation relating H and M .
b Given that a human with a body mass of 77kg has a heart mass of 0.462kg, determine k , the constant of proportionality.
c If a Chimpanzee has a body mass of 57kg, what is the heart mass H ?
d If a Macaque has a heart mass of 0.084kg, solve for its body mass M .

21 The blood circulation time C of a mammal is proportional to the fourth root of its body mass B .



- a** Using k as the constant of proportionality, form an equation relating C and B .
- b** Given that an elephant with a body mass of 5290 kg has a blood circulation time of 148 seconds, determine k , the constant of proportionality. Round your answer to one decimal place.
- c** If a Orangutang has a body mass of 40 kg, find C , its circulation time. Round your answer to one decimal place.
- d** If a Squirrel has a circulation time of 22 seconds, what is the body mass? Round your answer to one decimal place.